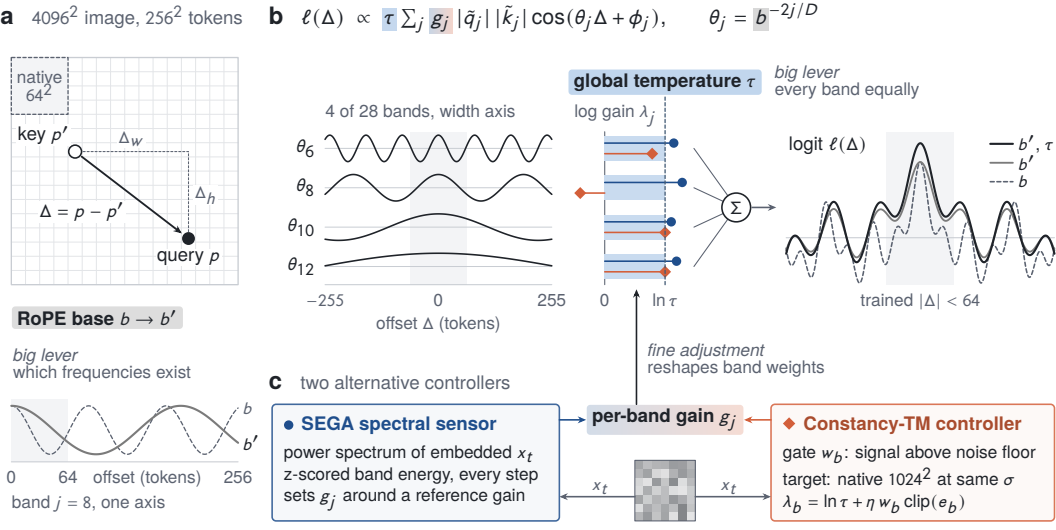
a
4096^2 image, 256^2 tokens
native 64^2
key p'
Δ_w
Δ_h
$\Delta = p - p'$
query p
b
$\ell(\Delta) \propto \tau \sum_j g_j |\tilde{q}_j||\tilde{k}_j| \cos(\theta_j \Delta + \phi_j), \quad \theta_j = b^{-2j/D}$
global temperature τ
big lever
every band equally
4 of 28 bands, width axis
log gain λ_j
logit $\ell(\Delta)$
b', τ
b'
b
θ_6
θ_8
θ_{10}
θ_{12}
Σ
−255
0
255
offset Δ (tokens)
0
$\ln \tau$
trained $|\Delta| < 64$
RoPE base $b \to b'$
big lever
which frequencies exist
fine adjustment
reshapes band weights
c
two alternative controllers
SEGA spectral sensor
power spectrum of embedded x_t
z-scored band energy, every step
sets g_j around a reference gain
per-band gain g_j
Constancy-TM controller
gate w_b: signal above noise floor
target: native 1024^2 at same σ
$\lambda_b = \ln \tau + \eta\, w_b\, \mathrm{clip}(e_b)$
x_t
x_t
b
b′
0
64
offset (tokens)
256
band $j = 8$, one axis